

# Fourier Transformation

## 4. Fourier Transformation Table of important functions

| Function $f(t)$                                                                 |                         | Fourier Transform $F(\omega)$                                |
|---------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------|
| $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$ |                         | $F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$ |
| • Time shifting                                                                 | $f(t-t_0)$              | $F(\omega) e^{-j\omega t_0}$                                 |
| • Frequency Shifting                                                            | $f(t) e^{-j\omega_0 t}$ | $F(\omega - \omega_0)$                                       |
| • Time Scaling                                                                  | $f(\alpha t)$           | $\frac{1}{ \alpha } F\left(\frac{\omega}{\alpha}\right)$     |
| • Symmetry                                                                      | $F(t)$                  | $2\pi f(-\omega)$                                            |
| • Derivative                                                                    | $\frac{d^n f(t)}{dt^n}$ | $(j\omega)^n F(\omega)$                                      |
| • Derivative                                                                    | $(-jt)^n f(t)$          | $\frac{d^n F(\omega)}{d\omega^n}$                            |

|                       |                                  |                                                                        |
|-----------------------|----------------------------------|------------------------------------------------------------------------|
| • Integral            | $\int_{-\infty}^t f(\tau) d\tau$ | $\frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$                   |
| • Dirac delta         | $\delta(t)$                      | 1                                                                      |
| • Dirac delta         | $e^{j\omega_0 t}$                | $2\pi\delta(\omega - \omega_0)$                                        |
| • Sgn function        | $\text{sgn}(t)$                  | $\frac{2}{j\omega}$ [F.T.]                                             |
| • Sgn function        | $\frac{j}{\pi t}$                | $\text{sgn}(\omega)$ [I.F.T.]                                          |
| • Unit-Step function  | $u(t)$                           | $\pi\delta(\omega) + \frac{1}{j\omega}$                                |
| • Cosine function     | $\cos(\omega_0 t)$               | $\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$           |
| • Sine function       | $\sin(\omega_0 t)$               | $\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$ |
| • Lorentzian Function | $e^{-\alpha t }$                 | $\frac{2\alpha}{\alpha^2 + \omega^2}$                                  |
| • Gaussian function   | $e^{-t^2/2\sigma^2}$             | $\sqrt{2\pi} \cdot \sigma e^{-\sigma^2\omega^2/2}$                     |

|                      |                                                         |                                                                                     |
|----------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------|
| •                    | $u(t)e^{-\alpha t}$                                     | $\frac{1}{\alpha + j\omega}$                                                        |
| • Heaviside function | $h(t) = \frac{1}{ t }$                                  | $\frac{2}{\omega} \sin\left(\frac{\omega}{2}\right) \{-\infty < + < \infty\}$       |
| •                    | $\int_{-\infty}^{\infty} e^{-j(\omega - \omega_0)t} dt$ | $2\pi\delta(\omega - \omega_0)$                                                     |
|                      | $\int_{-\infty}^{\infty} e^{-j2\pi(f - f_0)t} dt$       | $\delta(f - f_0)$                                                                   |
| • Heaviside function | $h(t)$                                                  | $\sum_{n=-\infty}^{\infty} \delta(t - nT)$ [F.T.]                                   |
| • Heaviside function | $H(f)$                                                  | $\frac{1}{T} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T}\right)$ [I.F.T.] |